

# Juan Carlos Cattaneo

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## Professional Summary

Software Engineer with 5+ years designing hardware-software integration systems and developer-facing APIs for medical devices. Currently leading the acquisition software team, driving OCT engine development, and cross-functional technical initiatives. Specialized in backend systems, platform APIs, and abstraction layers, with full-stack experience spanning C# and .NET on the backend to React and Next.js on the frontend. Proficient in C# and Azure DevOps, skilled at collaborating across functional teams to deliver scalable, well-tested solutions.

## Technical Skills

Languages: C# (Primary), Python, C++, JavaScript, Ruby

Frameworks & Libraries: .NET/WPF, React, Next.js, Flask, FastAPI, Ruby on Rails

Data: MySQL, SQLite, PostgreSQL, MongoDB, Supabase

Cloud + Tools: Azure DevOps, CI/CD, Git, Github, OpenCV, Ollama, Vercel, Agile / Scrum

Hardware: Cameras, Sensors, Motors, Arduino, Raspberry Pi, Jetson Nano

AI: Claude Code, GitHub Copilot, Ollama

## Professional Experience

**Software Engineer**   |   **Acquisition Team Lead**   |   **Carl Zeiss Meditec**

June 2021 - Present   |   Medical Device Software Development

- **Acquisition Team Lead:** Leading the acquisition software, driving technical direction for OCT engine development, and coordinating cross-functional efforts across subsystems.
- **AI-Assisted Development:** Integrated GitHub Copilot and Claude Code into daily workflows to accelerate development, code review, unit tests, and documentation.
- **Platform API Development:** Designed and maintained an OCT engine interface layer (DLLs/libraries) consumed by multiple teams for medical imaging hardware abstraction.
- **API Architecture & Integration:** Architected 30+ interface methods for camera, sensor, and motor systems. Refactored legacy code while preserving backward compatibility.
- **Platform Engineering:** Built C# WPF/MVVM solutions with 97%+ unit test coverage and architected CI/CD pipelines for automated build, test, and deployment.
- **Complex Systems Debugging:** Reduced critical defects by 40% through root cause analysis of multi-subsystem hardware-software integration issues.
- **Backend Systems Integration:** Built C++ sensor control backend and C# WPF frontend. Developed Linux/OpenCV applications for image processing workflows.
- **Code Quality & Refactoring:** Refactored 80% of legacy WPF codebase using OOP principles. Established MS Unit and NUnit testing frameworks.
- **Python Automation & Tooling:** Built Python APIs and automation tools for hardware validation and data analysis used by systems and research teams.

**Software Engineer Intern**   |   **Carl Zeiss Meditec**

Intern - Summer/Fall 2021   |   Medical Device Software Development

- **Front-End SWE:** Built React web app with Hooks. 90%+ unit test coverage.
- **SQA:** Automated desktop and web tests using Tosca and Testim IO.

## Education

|                                       |  |                                                 |      |
|---------------------------------------|--|-------------------------------------------------|------|
| California State University, East Bay |  | B.S. Business Administration - IT Management    | 2019 |
| App Academy Certificate               |  | Full-Stack Software Engineering (1,000+ hrs)    | 2020 |
| Diablo Valley College                 |  | Certificate - Computer Science - Program Design | 2023 |
| Diablo Valley College                 |  | Associate - Business Administration - Economics | 2015 |